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Original Article

An evolutionary constraint between reproduction and growth in a clade of *Miconia* (Melastomataceae)

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ABSTRACT

The reproductive allocation (RA) theory posits that investment in reproductive structures reduces resources for other plant body parts, imposing an intrinsic trade-off between sexual reproduction and vegetative growth. Under this theory, evolving greater reproductive structures would impose an evolutionary constraint upon vegetative growth, regardless of environmental conditions. However, this theoretical expectation has never been tested at the macroevolutionary scale until now. Based on herbarium specimens, we measured seed mass and specific leaf area (SLA), a proxy for vegetative growth, from *Miconia* species inhabiting rainforests and open-vegetation habitats in the Neotropics, representing different conditions for photosynthesis. We contrasted distinct scenarios of evolutionary correlation between seed mass and SLA values via phylogenetic comparative methods. We found similar support for two scenarios of evolutionary correlation. Under the most likely scenario, seed mass and SLA values evolved under a negative evolutionary correlation with steady magnitude in rainforest and open-vegetation habitats. Under the second-most likely scenario, seed mass and SLA values evolved under a negative evolutionary correlation only in open-vegetation habitats. The first scenario indicated that sexual reproduction imposed an evolutionary constraint upon plant growth regardless of environmental conditions, fully supporting the RA theory. Nonetheless, the second scenario highlights that environmental conditions can determine the emergence of photosynthate trade-offs within plant bodies, consequently changing the evolutionary correlation between sexual reproduction and growth. Regardless of environmental effects, our study indicates that sexual reproduction can impose evolutionary constraints to vegetative growth, thus impacting the evolution of plant economic strategies.

Keywords: economic strategy; evolutionary correlation; fast-slow spectrum; reproductive allocation; seed mass; SLA

INTRODUCTION

Plant lineages have specialized to habitats with distinct resource conditions (Woodward *et al.* 2004, Batjes 2014), ranging from resource-rich under competition to resource-limited due to stressors (He *et al.* 2013). Facing these challenges, plant lineages have evolved a continuum of traits that primarily allocate resources to either growth or defence, i.e. saving resources from herbivory and abiotic stress (Coley *et al.* 1985, Chapin *et al.* 1993). These resource allocation patterns, i.e. plant economic strategies, have been assumed to be adaptive responses to environmental resource conditions (Reich 2014), since growth and defence traits display broad-scale associations with climatic variables modulating resource assimilation by plants (e.g. Wright *et al.* 2005, Moles *et al.* 2007, 2009). However, reproductive traits have been marginally considered in the current theory on plant economic strategies (e.g. Saatkamp *et al.* 2019, Roddy *et al.* 2021), despite historical evidence of their resource costs for plants (Obeso 2002). Reproductive structures have been long assumed to demand far more resources than they can produce, thereby acting as resource sinks within plant bodies (Bazzaz *et al.* 1987). This resource-sink effect would occur from inflorescence to seed development, but much stronger in latter reproductive structures (Bazzaz *et al.* 1987). Hence, sexual reproduction would be inherently resource demanding for plants, likely imposing resource trade-offs with

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other body functions (Obeso 2002). Despite these expectations, macroevolutionary studies have seldom investigated how resource allocation to sexual reproduction evolves with plant economic strategies (Johnson *et al.* 2015), often emphasizing on the dilemma of growth and defence (Züst and Agrawal 2017).

Under the RA theory (Karlsson and Méndez 2005), sexual reproduction constrains vegetative growth by decreasing plants' photosynthetic capacity while increasing the demand for photosynthates. Initially, inflorescence development reduces the photosynthetic area of plant bodies by producing non-leafy or non-green structures (Watson and Casper 1984, Aschan and Pfanz 2003). Later, flowers withdraw photosynthates to produce secondary metabolites for pollinator attraction (Paul *et al.* 2022), further expending photosynthates to produce nectar or other rewards for pollinators (e.g. Southwick 1984, Ashman and Schoen 1997). Finally, seeds withdraw photosynthates to produce starch and other reserves for future offspring (Borisjuk *et al.* 2004), while having a minimal photosynthesis (Watson and Casper 1984, Aschan and Pfanz 2003). Hence, sexual reproduction imposes a progressive resource-sink effect that limits photosynthates to plant growth. For instance, the annual *Abutilon theophrasti* Medik. (Malvaceae) accumulates a smaller body biomass when producing larger seeds across different soil conditions (Sugiyama and Bazzaz 1998), highlighting a trade-off between sexual reproduction and vegetative growth regardless of edaphic conditions. Moreover, the perennial *Plantago major* L. (Plantaginaceae) accumulates smaller body biomass when producing larger seeds in previous years (Reekie *et al.* 2002), indicating a long-term effect of sexual reproduction upon vegetative growth. Following this perspective, evolving greater reproductive structures would enhance the resource limitations for plant growth (Lovett Doust 1989), forecasting a negative evolutionary correlation (Arnold *et al.* 2001). However, such an evolutionary correlation presumes an inherent limitation of photosynthates within plant bodies (Tuomi *et al.* 1983), disregarding environmental conditions modulating photosynthesis. Thus, sexual reproduction and plant growth can experience a negative evolutionary correlation whose emergence might depend on environmental resource conditions.

The Neotropics holds one third of the world's plant species diversity across distinct habitat types (Antonelli and Sanmartín 2011), ranging from rainforests to different open-vegetation habitats, such as seasonally dry shrublands, savannas, montane grasslands, and deserts (Hughes *et al.* 2013). The rainforests provide high precipitation conditions (~1500 mm/year) for plants (Jaramillo 2023), sustaining a great primary productivity (Raich *et al.* 1991). In contrast, open-vegetation habitats often impose water limitation to plants via droughts and disturbance regimes, such as fire and frost (Jaramillo 2023), limiting primary productivity below levels of rainforests (Raich *et al.* 1991). Nonetheless, water availability is the main limiting factor across these vegetation types in the Neotropics (Segovia *et al.* 2020), also controlling biomass stocks in rainforests (Poorter *et al.* 2017). Hence, plant lineages can face distinct conditions of water availability across neotropical habitats, forecasting the evolution of plant economic strategies. For instance, in a neotropical group of *Chamaecrista* (L.) Moench (Fabaceae), species evolved greater resource-storage capacity in the *campos rupestres* where dry seasons can constrain plant carbon fixation (Prochazka *et al.* 2024). Moreover, plant lineages also evolve different reproductive traits across neotropical habitats. For instance, in the neotropical tribe Myrtaeae (Myrtaceae), species evolved relatively larger inflorescences in savannas than in rainforests (Vasconcelos *et al.* 2019). Thus, neotropical plant lineages are suitable models to investigate the evolutionary patterns of reproductive allocation under distinct environmental resource conditions.

Our study assessed whether sexual reproduction can impose evolutionary constraints to vegetative growth and, consequently, impact plant economic strategies. Based on the RA theory, we hypothesized that evolving greater reproductive structures would constrain vegetative growth due an intrinsic trade-off upon photosynthates within plant bodies, regardless of environmental conditions impacting photosynthesis. To address this hypothesis, we measured seed mass and specific leaf area (SLA) from *Miconia* Ruiz & Pav species, assuming the SLA as a proxy for plant growth (Reich *et al.* 1998). We measured these traits on herbarium specimens representing *Miconia* species that inhabit rainforests and open-vegetation habitats in the Neotropics, thus representing different conditions for photosynthesis. We then tested patterns of evolutionary correlation between seed mass and SLA values via phylogenetic comparative methods. In support of our hypothesis, we would expect that (i) seed mass and SLA values should have evolved under a negative evolutionary correlation and that (ii) such a negative evolutionary correlation should have been steady across rainforest and open-vegetation habitats.

MATERIAL AND METHODS

Unless otherwise stated, we performed data analyses in the R environment v.4.1.0 (R Development Core Team 2021). The R code and datasets are available at https://github.com/eknery/reproductive_allocation_Neotropics. Throughout, we only cited the main functions and their respective R packages.

Study area

The study area encompassed the extra-Andean Neotropics, mostly the phytogeographic domains of the Amazon, Atlantic Forest, Cerrado, and Caatinga. These phytogeographic domains are variable with respect to precipitation and soil characteristics, holding habitats with distinct resource conditions for plants. The Amazon is mostly under a tropical climate with monsoons and without dry seasons (Alvares *et al.* 2013). The Amazon holds a mosaic of soil types across its river systems where soil-nutrient availability overall decreases in white-sand and floodplain areas (Quesada *et al.* 2020). The Atlantic Forest is mostly under a humid subtropical climate, with dry seasons in some inland regions during the winter or summer (Alvares *et al.* 2013). The Atlantic Forest has a greater edaphic variation across its elevational gradient where soil-nutrient availability decreases towards the lowland coastal dunes and the montane rock outcrops (Cunha *et al.* 2019). The Cerrado is mostly under a tropical climate with dry winters (Alvares *et al.* 2013), that promote fire regimes over most of its extension (Oliveira *et al.* 2022). The Cerrado mostly holds nutrient-poor soils due to acidic pH and high aluminium concentration (Werneck 2011). The Caatinga is mostly under a semi-arid climate that can impose prolonged droughts (Alvares *et al.* 2013), but it overall holds nutrient-rich soils (Werneck 2011).

Focal biological group

We utilized a major clade of *Miconia* Ruiz & Pav to address our question and hypothesis. *Miconia* is a hyperdiverse plant genus (~2000 spp.) restricted to the Neotropics (Moonlight *et al.* 2024), occupying both rainforest and open-vegetation domains (Michelangeli *et al.* 2022). *Miconia* species can be shrubs, treelets, or trees that often are pioneers in community succession (Michelangeli *et al.* 2004, Goldenberg *et al.* 2008). These species are primarily bee-pollinated but often perform self-pollination and apomixis, assuring reproduction even without pollinators (Renner 1989, Dellinger *et al.* 2022). After pollination or apomixis, these species produce small berries that are dispersed by various bird species and some insect groups, thus holding a great dispersal potential (Messeder *et al.* 2021). Hence, *Miconia* species can disperse to different habitat types, but their persistence still depends on their economic strategy.

Our focal group was the *Miconia* supersect. *Discolores* Caddah & Goldenberg (Caddah *et al.* 2022). This clade comprises about 77 species (Caddah *et al.* 2022), of which 66 (85%) have known phylogenetic placement (Nery *et al.* 2023). In this *Miconia* clade, species can occupy rainforests and open-vegetation habitats in the Neotropics. Species specialized in rainforests are mostly found in the Amazon and Atlantic Forest domains, whereas species specialized in open-vegetation habitats are mostly found in the Cerrado domain (Goldenberg *et al.* 2024). Species occupying both rainforests and open-vegetation habitats are often found in more than one phytogeographic domain (Goldenberg *et al.* 2024). Hence, species from our focal *Miconia* group have faced distinct conditions for photosynthesis, perhaps experiencing carbon trade-offs among their body functions.

Phylogenetic hypotheses

To allow the inference of macroevolutionary processes, we utilized a sample of phylogenetic hypotheses for our focal *Miconia* group. We retrieved a random sample ($N = 100$) of phylogenetic trees from the posterior distribution inferred by Nery *et al.* (2023). Their phylogenetic analysis was the most comprehensive (66 spp.) for our focal group, utilizing two nuclear and four plastidial loci and considering two calibration points for divergence time estimates (Nery *et al.* 2023). This sample of phylogenetic trees allowed an explicit incorporation of phylogenetic uncertainty in our evolutionary analyses (Rangel *et al.* 2015). We also retrieved a Maximum Clade Credibility (MCC) tree to illustrate our results.

Characterizing resource conditions via habitat type

To characterize the resource conditions faced by our focal *Miconia* species, we utilized broad categories of habitat type. We surveyed which habitats are occupied by our focal *Miconia* species by consulting the *Flora e Funga do Brasil* database (Goldenberg *et al.* 2024), and non-published monographs of the *Miconia* supersect. *Discolores.* The survey returned 16 habitats that were cited for a different number (N) of species: the Atlantic Forest's rainforest (*floresta ombrófila densa*, $N = 38$), the Amazonian upland rainforest (*floresta de terra firme*, $N = 17$), the Cerrado's savanna (*cerrado s.s.*, $N = 16$), the Cerrado's riparian forest (*floresta ciliar*, $N = 11$), the Amazonian white-sand woodland (*campinarana*, $N = 10$), the Amazonian savanna (*savana amazônica*, $N = 5$), the Atlantic Forest's semi-deciduous forest (*floresta estacional semedicidual*, $N = 5$), the Atlantic Forest's montane rocky outcrops (*afloramento rochosos*, $N = 5$), the Cerrado's rocky grasslands (*campos rupestres*, $N = 5$), the Amazonian floodplain rainforest (*floresta de várzea*, $N = 3$), the Atlantic Forest's Araucaria rainforest (*floresta ombrófila mista*, $N = 2$), the Atlantic Forest's coastal woodlands (*restinga*, $N = 2$), the Atlantic Forest's deciduous forest (*floresta estacional decidual*, $N = 2$), the Amazonian floodplain grassland (*campo de várzea*, $N = 1$), the Caatinga's seasonally dry forest (*caatinga*, $N = 1$), and the Caatinga's white-sand woodland (*carrasco*, $N = 1$).

We grouped these 16 habitats into three categories based on their physical structure, water availability, and disturbance regimes: the rainforest category included vegetations with a closed canopy and without pronounced droughts or disturbances (the Amazonian upland rainforest + the Atlantic Forest's rainforest and Araucaria forest + the Cerrado's riparian forest); the transition-habitat category included vegetations with a closed canopy, but with pronounced droughts or disturbances (the Amazonian floodplain rainforest + the Atlantic Forest's deciduous and semi-deciduous forests + the Caatinga's seasonally dry forest); the open-vegetation category included vegetations with an open canopy and with pronounced droughts or disturbances (the Amazonian savanna, white-sand woodland, and floodplain grasslands + the Atlantic Forest's coastal woodland and montane rock outcrops + the Cerrado's savanna and rocky grasslands + the Caatinga's white-sand woodland). Hence, these habitat categories represent decreasing conditions of water availability, from rainforests to transition and open-vegetation habitats, respectively.

Based on these habitat categories, we delimited three character states of habitat type for our focal *Miconia* species: 'rainforest' for species specific to the rainforest category; 'open-vegetation' for species specific to the open-vegetation category; and 'generalist' for species found in more than one habitat category. There were no species specific to the transition-habitat category. We did not define character states based on the 16 habitats in our literature survey for different reasons. First, our objective was inferring the evolution of reproductive allocation and growth in distinct resource conditions, rather than inferring evolution in each habitat. Second, some habitats were quite similar (e.g. the Amazonian savanna and white-sand woodland), being better interpreted as a single habitat category. Finally, some habitats were occupied just by one species, preventing a statistical replication for macroevolutionary analyses.

Characterizing reproductive allocation and vegetative growth

The dry mass of reproductive and vegetative structures has been traditionally interpreted as carbon allocation to their underlying functions (Reekie and Bazzaz 1987a), albeit other resources can follow other patterns (Reekie and Bazzaz 1987b). Hence, we sought to measure dry-mass traits from herbarium specimens to characterize the reproductive allocation and vegetative growth in our focal *Miconia* species. We measured dry-mass traits from specimens held by five herbaria: FLOR, MBML, MG, NY, and UPCB (Thiers 2023). We sampled 205 specimens representing 58 species (3 ± 1 specimens per species), including specimens from different localities. We calculated the median value of conspecific specimens to represent species' phenotype.

Reproductive allocation refers to the investment of plant resources in the production of reproductive organs and associated structures, from inflorescences to seeds (Karlsson and Méndez 2005). However, in our focal *Miconia* group, most reproductive structures had a similar size or quantity across habitat types, preventing their use to measure changes in reproductive allocation. Inflorescence size varied little between rainforest (mean ± SD = 12.3 ± 4.8 cm) and open-vegetation (10.2 ± 5.9 cm) species ($F_{2,55}$ = 0.673, p-value = 0.515) (Supporting Information, Fig. S1A). Based on data by Nery *et al.* (2024), flower size was very similar between rainforest (3.7 ± 1.3 mm) and open-vegetation (3.7 ± 1.9 mm) species ($F_{2,55}$ = 0.246, p-value = 0.783) (Supporting Information, Fig. S1B). Also, fruit size was similar between rainforest (3.5 ± 1.2 mm) and open-vegetation (3.4 ± 0.9 mm) species ($F_{2,55}$ = 0.191, p-value = 0.826) (Supporting Information, Fig. S1C). Finally, the number of seeds per fruit also varied little between rainforest (18 ± 16) and open-vegetation (22 ± 15) species ($F_{2,55}$ = 0.30, p-value = 0.742) (Supporting Information, Fig. S1D).

In contrast with other reproductive structures, seed mass was variable between rainforest and open-vegetation species in our focal *Miconia* group. We measured seed mass (mg) by weighing dry fruits and counting their seed number, later dividing the whole-fruit dry mass by its seed number. Our measurement directly indicated the average seed mass per fruit, but it also indirectly indicated the total seed mass per specimen. First, seed mass was positively correlated with fruit size across specimens (slope = 0.51, R^2 = 0.19, p-value < 0.001; Supporting Information, Fig. S2A), indicating that heavier seeds were produced in larger fruits. Second, fruit size was uncorrelated with inflorescence size across specimens (p-value = 0.660; Supporting Information, Fig. S2B), countering a possible compensation of larger fruits by shorter inflorescences with fewer fruits. Third, fruit size was also uncorrelated with plant size across specimens (p-value = 0.653; Supporting Information, Fig. S2C), countering a possible compensation of larger fruits by shorter plants with fewer fruits. Hence, heavier seeds were produced in larger fruits, but without a reduction in fruit number via shorter inflorescences or shorter plants, thus indicating an increase in total seed mass per specimen. Therefore, we utilized seed mass as a proxy for reproductive allocation.

To represent vegetative growth, we measured the SLA of herbarium specimens. SLA values directly correlate with photosynthetic rates (Reich *et al.* 1998), also directly determining relative growth rates (Poorter *et al.* 2009). Consequently, SLA values are variable over water availability gradients, reducing towards drought-prone and less-productivity habitats (Poorter *et al.* 2009). We measured SLA (mm^2/mg) by collecting and weighing a 10-mm^2 wide tissue sample from the medial leaf blade of herbarium specimens.

It is worth noting that plant traits measured from herbarium specimens may bear measurement errors (Heberling 2022). However, our main objective was investigating the correlated evolution of traits in species occupying distinct habitat types, rather than estimating fine differences of economic strategies within the same species or habitat. Still, herbarium-derived trait values tightly correlate with field-derived trait values (e.g. Perez *et al.* 2020), providing a rough approximation of plant economic strategies.

Inferring the evolution of habitat type

To infer the past resource conditions faced by our focal *Miconia* group, we performed an ancestral reconstruction of habitat type. We considered habitat type a categorical niche dimension and modelled its evolution via transition rate models (Joy *et al.* 2016). We considered three character states of habitat type and consequently allowed for six transition rates: 'rainforest' to 'generalist' (q12), 'rainforest' to 'open-vegetation' (q13), 'generalist' to 'rainforest' (q21), 'generalist' to 'open-vegetation' (q23), 'open-vegetation' to 'rainforest' (q31), and 'open-vegetation' to 'generalist' (q32).

We fitted five transition models to the character states of habitat type: an equal rates (ER) model, where all transition rates had a single value; an unordered and symmetrical (UNSY) model, where transition rates towards and away from a same character state had a single value; an unordered and asymmetrical (UNAS) model, where transition rates towards and away from each character state had a distinct value; an ordered and symmetrical (ORSY) model, where transition rates towards and away from a same character state had a single value, but disabling transition rates between 'rainforest' and 'open-vegetation' states (q13 = q31 = 0); and an ordered and asymmetrical (ORAS) model, where transition rates towards and away from each character state had a distinct value, but disabling transition rates between 'rainforest' and 'open-vegetation' states (q13 = q31 = 0). The unordered models represent scenarios where habitat type evolves rapidly, allowing direct changes from the 'rainforest' and 'open-vegetation' states. In contrast, the ordered models represent scenarios where habitat type evolves more gradually, requiring the 'generalist' state as an intermediate between the 'rainforest' and 'open-vegetation' states.

We fitted all transition rate models to our sample of phylogenetic trees (N = 100), evaluating their goodness of fit by the corrected Akaike Information Criterion (AICc) (Sugiura 1978). We assumed a cut-off of two AICc scores to choose a more complex model over a simpler model. We chose the transition matrix with the best fit across most phylogenetic trees. Based on the best-fit transition matrix, we conducted an ancestral reconstruction of habitat type by utilizing the stochastic character mapping (SIMMAP) method (Bollback 2006), as implemented in the 'phytools' package (Revell 2012). We simulated and summarized 100 characters maps for each phylogenetic tree, assuming the 'generalist' state as more likely (p = 0.5) for the last common ancestor. Such a prior derives from a previous macroevolutionary study that indicated a wider niche breadth for the last common ancestor of our focal *Miconia* group (Nery *et al.* 2023). We utilized the posterior probabilities to assign ancestral species to a character state of habitat type.

Testing the evolutionary correlation between seed mass and SLA

To test whether reproductive allocation and vegetative growth have jointly evolved in our focal *Miconia* group, we fitted and contrasted different models of evolutionary correlation to seed mass and SLA values. We utilized evolutionary rate models to infer patterns of evolutionary correlation between traits (Revell and Collar 2009), as implemented in the function *evolvcv.lite* from the R package 'phytools' (Revell 2012). This model represents the evolution of two or more traits by stochastic changes drawn from

a multivariate normal distribution whose variance and covariance are proportional to the divergence time from a phylogenetic tree (Revell and Collar 2009). The variance parameters represent the evolutionary rates of traits whereas the covariance parameter reflects their evolutionary correlation (Revell and Collar 2009). Variance and covariance parameters can be modelled as either fixed or variable with regard to a discrete character mapped onto a phylogenetic tree, thus representing scenarios where trait variation and co-variation are either independent or dependent on other variables, respectively (Revell *et al.* 2022).

Based on the above framework, we considered seven models of evolutionary rates for seed mass and SLA values: model 1, fixed variance for both seed mass and SLA and fixed covariance across habitat types; model 2, variable variance for both seed mass and SLA and fixed covariance across habitat types; model 3a, fixed variance for both seed mass and SLA and variable covariance across habitat types; model 3b, fixed variance for seed mass, variable variance for SLA, and variable covariance across habitat types; model 3c, variable variance for seed mass, fixed variance for SLA, and variable covariance across habitat types; model 4, variable variance for both seed mass and SLA and variable covariance across habitat types. The models with fixed covariance (models 1 and 2) represent an evolutionary correlation independent of habitat type, whereas the models with variable covariance (models 3a, 3b, 3c, and 4) represent an evolutionary correlation dependent on habitat type. We employed different versions of model 3 (a, b, and c) to accommodate rate heterogeneity across habitat types, which can interfere with the detection of correlated evolution (Revell *et al.* 2022).

Before model fitting, we log-transformed seed mass and SLA values to better conform to a multivariate normal distribution (Revell and Collar 2009). We fitted all evolutionary rate models to our sample of phylogenetic trees ($N = 100$), evaluating their goodness of fit by the Akaike Information Criterion (AIC) (Akaike 1974). The AIC scores have shown the lowest type I error with the greatest statistical power for choosing evolutionary rate matrices (Revell and Collar 2009). The model selection followed the criteria previously described for transition rate models. We summarized parameter estimates based on the best-fit evolutionary rate model, transforming covariance estimates to correlation values for convenience.

RESULTS

Transition rates differed in rainforest and open-vegetation habitats

Our focal *Miconia* group included species inhabiting rainforests ($N = 29$), open-vegetation habitats ($N = 11$), and both habitat types (generalists, $N = 15$). The ORSY model of transition rates was the best-fit model for habitat types across all phylogenetic trees. Under the ORSY model, transition rates towards and away from the generalist state were symmetrical (q12 = q21 and q23 = q32). Nonetheless, transition rates were three times greater between the rainforest and generalist states [median ± interquartile range (iqr) = 0.62 ± 2.20 transition per Mya] than between the open-vegetation and generalist states (0.22 ± 0.07 transition per Mya).

Based on the ORSY model, our focal *Miconia* group originated from a generalist ancestor, about 8.3 Mya (Fig. 1). This ancestor diversified and gave rise to rainforest lineages at three different times. These rainforest lineages evolved back to generalist lineages at 10 different times. Some generalist lineages later gave rise to open-vegetation lineages at six different times, from 2 Mya to recently.

Seed mass and SLA values varied more in rainforest than in open-vegetation habitats

Seed mass and SLA values varied across habitat types (Fig. 2). Seed mass was 42 to 80% greater for rainforest species (median ± iqr: 1.50 ± 2.33 mg) than for generalist (1.05 ± 2.03 mg) and open-vegetation species (0.83 ± 0.51 mg), respectively. SLA was 25 to 75% greater for rainforest species (9.80 ± 5.00 mm2/mg) than for generalist (7.78 ± 4.25 mm2/mg) and open-vegetation species (5.60 ± 2.25 mm2/mg), respectively.

Seed mass and SLA held a negative evolutionary correlation

Two models of evolutionary rates had similar goodness-of-fit to seed mass and SLA values. The model 1 was the best-fit for 37% of phylogenetic trees, followed by the model 3c for 34% of phylogenetic trees. This pattern suggests that both models had similar explanatory power over our data when considering phylogenetic uncertainty. Hence, we reported estimates from both models, further discussing their implications.

Under model 1, seed mass had greater evolutionary rates [median ± iqr: 0.32 ± 0.07 ln(mg)/million years] than SLA values [0.07 ± 0.01 ln(mm/mg)/million years]. Moreover, seed mass and SLA held negative values of evolutionary correlation (−0.27 ± 0.05), but without changes across habitat types (Fig. 3A–B). This pattern supports that evolving a greater reproductive allocation can constrain vegetative growth regardless of resource conditions.

Under model 3c, seed mass had constant evolutionary rates across habitat types [median ± iqr: 0.41 ± 0.13 ln(mg)/million years], but SLA had decreasing evolutionary rates [ln(mm/mg)/million years] from the generalist (0.144 ± 0.104), to the rainforest (0.041 ± 0.041), and open-vegetation (0.009 ± 0.004) habitats. Moreover, seed mass and SLA held decreasing values of evolutionary correlation from rainforest (0 ± 0.30), to generalist (−0.64 ± 0.28), and open-vegetation habitats (−0.99 ± 0.12) (Fig. 3C–D). This pattern supports that evolving a greater reproductive allocation can constrain vegetative growth, but such an evolutionary constraint becomes pronounced only in water-limited habitats.

DISCUSSION

Under the most likely scenario (model 1), our focal *Miconia* species evolved seed mass and SLA under a negative correlation whose magnitude was steady between rainforest and open-vegetation habitats, thus supporting our predictions. This pattern indicates that sexual reproduction imposes an evolutionary constraint upon vegetative growth. The RA theory predicts that reproductive structures compromise vegetative growth by reducing a plant's photosynthetic capacity while increasing the demand for photosynthates (Karlsson and Méndez 2005). For instance, in the annual *Ambrosia trifida* L. (Asteraceae), flowers

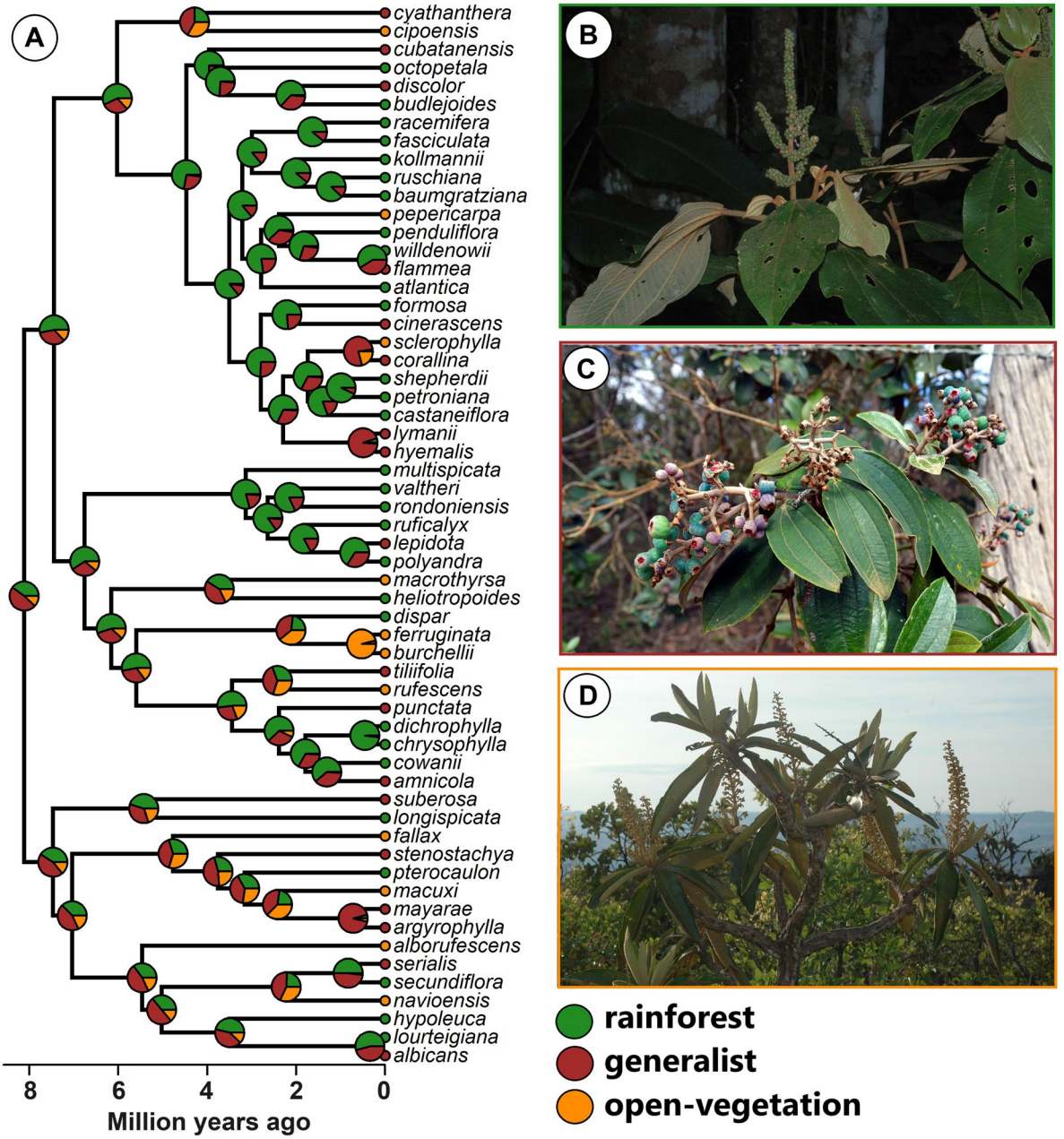


Figure 1. Phylogenetic relationships and habitat reconstruction in the *Miconia* supersect. *Discolores*. A, current and ancestral habitat types considering a SIMMAP reconstruction over the MCC tree. B, *Miconia petroniana* Cogn. & Saldanha, a forest species. C, *Miconia albicans* (Sw.) Steud., a generalist species. D, *Miconia ferruginata* DC., an open-vegetation species. Pictures by Renato Goldenberg and Julia Meirelles.

and seeds fix less than 50% of the carbon needed for their own development and maintenance (Bazzaz and Carlson 1979), highlighting an inherent resource-sink effect from reproductive structures. Moreover, in the perennial *Pinguicula vulgaris* L. (Lentibulariaceae), fruiting reduces almost 50% of the vegetative bud mass produced in the following growing season (Worley and Harder 1996), indicating that reproductive structures primarily withdraw resources from vegetative growth. Based on these patterns, evolving greater reproductive structures would require enhancing traits that compromise the vegetative growth. To our knowledge, our study is the first to infer that sexual reproduction can impose evolutionary constraints to vegetative growth, supporting the RA theory at the macroevolutionary scale.

Plants have been long assumed to have a limited carbon budget for their body functions (Mooney 1972). This intrinsic carbon limitation would have imposed selective pressures favouring specific trait combinations that could maintain a minimal net carbon gain (Mooney 1972), thus prompting the evolution of plant economic strategies. Multiple reproductive traits, from inflorescences to seeds, would have also been selected to maintain plant economic strategies (Lovett Doust 1989). Hence, the evolution of reproductive structures would not only impose constraints but also become constrained under the carbon trade-off of plant bodies. Under this premise, the negative evolutionary correlation between seed mass and SLA could also be interpreted as an evolutionary constraint of vegetative functions upon sexual reproduction. Such an

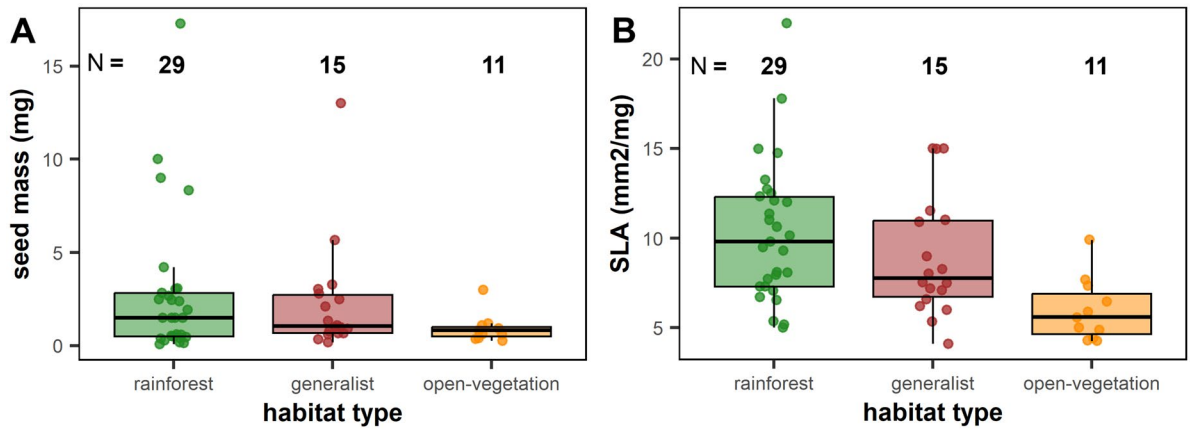


Figure 2. Seed mass and SLA values of *Miconia* species and their sample size (N) across habitat types. Each point represents a species' mean phenotype. Boxes represent the interquartile ranges, while the vertical lines represent 1.5 times the interquartile ranges. Black horizontal lines indicate the median.

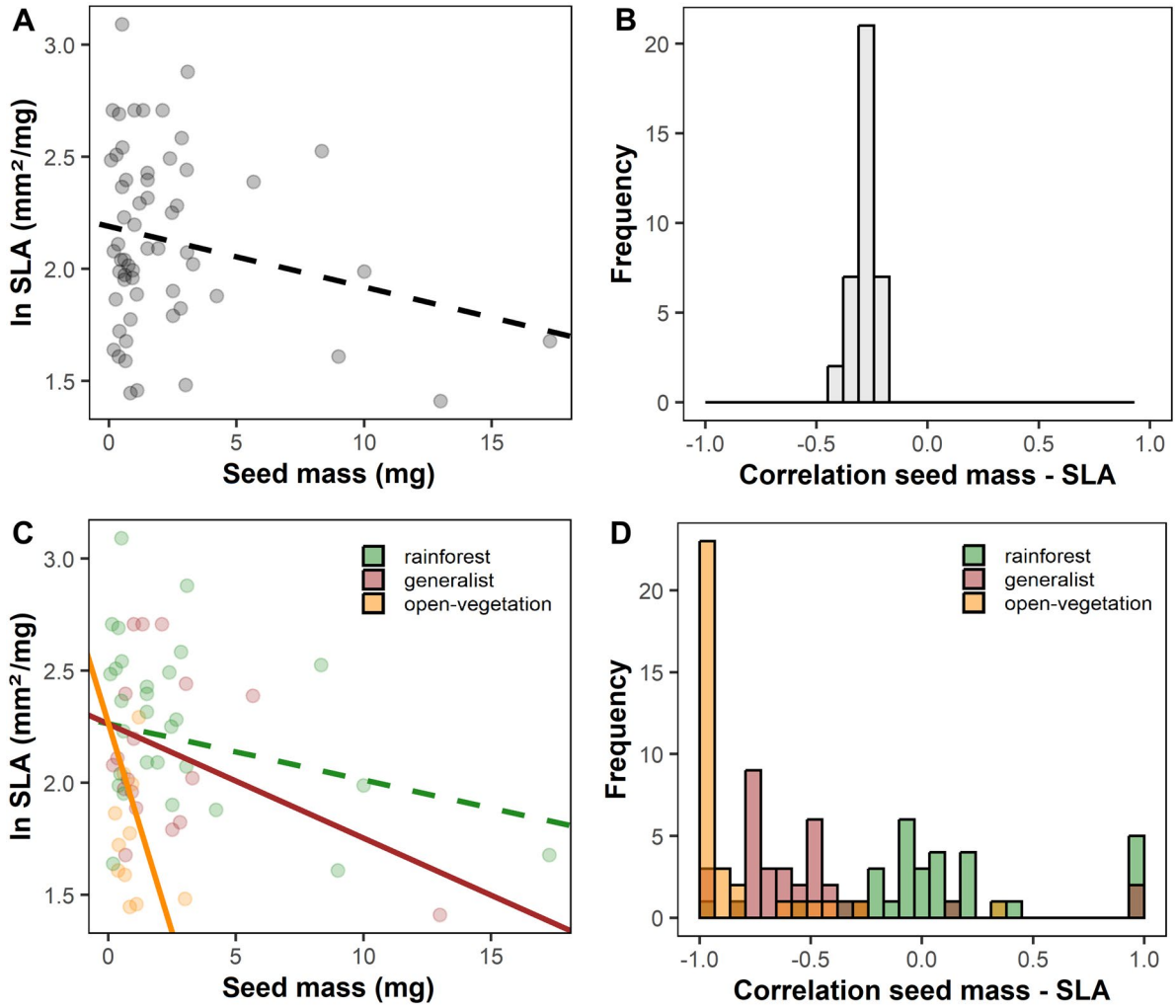


Figure 3. Relationship and evolutionary correlation between seed mass and SLA values in the *Miconia* supersect. *Discolores*. A, relationship between traits disregarding habitat type. B, estimates of evolutionary correlation based on model 1, disregarding habitat type. C, relationship between traits considering habitat type, each represented by a colour. D, estimates of evolutionary correlation based on model 3c, considering habitat type, each represented by a colour. Throughout, solid lines represent significant relationships inferred by PGLS models (Garland *et al.* 1993), while dashed lines represent non-significant relationships. PGLS models were used only to illustrate the current relationships between traits.

evolutionary constraint has been primarily supported by the smaller seed reproduction in clonal than in strictly sexual plants (Herben *et al.* 2015). Interestingly, many *Miconia* species can perform clonal reproduction via apomixis (Renner 1989, Caetano and Oliveira 2022), often producing low-quality pollen (Caetano *et al.* 2013), possibly reallocating resources from sexual reproduction to vegetative functions. Assuming this resource reallocation, apomixis could have been selectively favoured in *Miconia* species since it would have allowed increasing the vegetative function over sexual reproduction, an untested hypothesis so far. Hence, our results may reflect a reciprocal evolutionary constraint between sexual reproduction and vegetative growth, yet the effects of vegetative functions require further investigations.

Nonetheless, under the second-most likely scenario (model 3c), the negative evolutionary correlation between seed mass and SLA had different magnitudes in rainforest and open-vegetation habitats. More specifically, the negative evolutionary correlation was absent in rainforests and strong in open-vegetation habitats, where photosynthesis is limited by drought (Raich *et al.* 1991). This pattern indicates that carbon trade-offs between sexual reproduction and vegetative growth depend on environmental conditions influencing carbon uptake. The resource availability hypothesis has long stated that environmental resource conditions would select plant trait combinations that enhance either growth or defence (Coley *et al.* 1985). Later large-scale studies have indicated that patterns of plant trait correlation depend on environmental conditions determining photosynthetic rates (e.g. Wright *et al.* 2005). Hence, some patterns of plant trait correlation may only evolve under environmental conditions limiting resource uptake (Reich 2014). In this sense, our second-most likely scenario supports that environmental conditions determine whether resource trade-offs will impose selective pressures upon reproductive and vegetative traits together, yet with a smaller statistical support. A greater sample of plant lineages would have likely increased the statistical support for this second evolutionary scenario (Martin *et al.* 2023), since similar scenarios have been inferred for larger plant lineages. For instance, in the neotropical tribe Myrtaeae, species evolved relatively larger inflorescences in savannas than in rainforests, suggesting a greater resource allocation to sexual reproduction over growth in water-limited habitats (Vasconcelos *et al.* 2019). Hence, the negative evolutionary correlation between sexual reproduction and vegetative growth can be enforced under lower-photosynthesis conditions, highlighting the abiotic environment as a major selective pressure upon plant economic strategies.

Seeds require a great carbon input (Weber *et al.* 1997), while having a very low photosynthetic capacity (Watson and Casper 1984), thus representing a net resource allocation to reproduction. However, reproductive allocation can be measured by different methods, each of them with their limitations (Poorter and Sack 2012). In our study, we measured reproductive allocation as the average seed mass per fruit, which was indirectly associated with the total seed mass per specimen (see explanation in the Material and Methods). Another method would be estimating the total seed mass per specimen as a function of their inflorescence and plant sizes, i.e. an allometric method (Poorter and Sack 2012). This allometric method would require a validation of model estimates by empirical measures of seed production, possibly implying the extrapolation of allometric curves from well-collected to lesser-known species. Moreover, our measurement of seed mass primarily reflects carbon allocation to reproduction (Reekie and Bazzaz 1987a), leaving other resources types unmeasured (Reekie and Bazzaz 1987b). For instance, some habitat types in the Cerrado domain are highly phosphorus-limited (Abrahão *et al.* 2019), and this low phosphorous availability can decrease mineral nutrition to leaves (Oliveira *et al.* 2015), perhaps imposing a mineral trade-off between leaves and seeds. Measuring phosphorus in our focal *Miconia* species could unveil additional constraints between sexual reproduction and vegetative growth, an exciting possibility for further studies. Despite method limitations, our results represent the first evidence that reproductive allocation can impact plant economic strategy at the macroevolutionary scale.

Sexual reproduction has been marginally addressed by the current theory on plant economic strategies (Saatkamp *et al.* 2019, Roddy *et al.* 2021), despite past empirical studies indicating an effect of sexual reproduction upon vegetative growth (Obeso 2002). On the one hand, plant economic strategies have been interpreted as evolutionary responses to developmental and resource trade-offs between growth and defence (Herms and Mattson 1992). Under this dichotomous perspective, sexual reproduction and vegetative growth have been jointly analysed as a single proxy for plant fitness that can decrease owing to the costs of defence (Koricheva 2002). More recent theoretical frameworks on the growth-defense dilemma had acknowledged that sexual reproduction can compete for resources within plant bodies, but without explicitly proposing mechanistic explanations (Züst and Agrawal 2017). On the other hand, sexual reproduction has been considered a separated component of plant strategies along with growth and defence (Grime 1977), also evolving in response to resource availability and disturbance (Deyn 2017, Liu *et al.* 2025). Supporting this tripartite perspective, our study indicates that seed mass evolves with growth-related traits, likely under the effect of environmental resource conditions. Further macroevolutionary studies could reveal additional relationships between reproductive and vegetative traits. For instance, flowers and leaves could compete for common secondary metabolites and undergo mutual evolutionary constraints (Johnson *et al.* 2015). Thus, we may unveil a triad of resource trade-offs among plant functions, providing more comprehensive explanations on how plant economic strategies evolve.

CONCLUSION

The RA theory has long stated that sexual reproduction withdraws resources and limits the vegetative growth of plants. Still, sexual reproduction has been marginally considered in the current theory of plant economic strategies. Here we investigated whether sexual reproduction and vegetative growth have jointly evolved due to an intrinsic resource trade-off within plant bodies. We found that a negative evolutionary correlation between seed mass and SLA, a proxy for vegetative growth, in *Miconia* species inhabiting distinct neotropical habitats. This first pattern indicates that sexual reproduction imposes an evolutionary constraint upon vegetative growth regardless of environmental conditions, fully supporting the RA theory at a macroevolutionary scale. Nonetheless, we also found that seed mass and plant growth may evolve under a negative evolutionary correlation between only in open-vegetation habitats,

where resources are more limited. This second pattern highlights that environmental conditions can determine resource trade-offs within plant bodies, thus impacting on evolutionary correlations between sexual reproduction and vegetative growth. Despite alternative patterns, our study provides some evidence that sexual reproduction can be a costly function and consequently impacts the evolution of plant economic strategies.

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SUPPORTING INFORMATION

Supplementary data is available at *Botanical Journal of the Linnean Society* online.

CONFLICT OF INTEREST

None declared.

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DATA AVAILABILITY STATEMENT

The data underlying this article will be shared on reasonable request to the corresponding author: https://github.com/eknery/reproductive_allocation_Neotropics.

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